

Workflow and architecture patterns for working together 1: Workflows

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Outline

1 Current challenges and their canonical solutions

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- 2 DevOps 201
 - Communication
 - Throughput

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 - CI Branching patterns
 - Trade-offs
 - Enablers
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 - Definition
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Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks
- coupling of all services at the database layer
 - high storage costs
 - single point of failure risk
 - risk of cross-project interference

Canonical solutions to these challenges

- multiple teams → inverse Conway maneuver
- collaboration → business stream-aligned teams over activity oriented teams
- late integration → adopting a true continuous integration workflow
- database coupling → Bezos API mandate/“Microservices”

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1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' *Datamation*, 1967

Conway's law: examples

- 'If you have four groups working on a compiler, you'll get a 4-pass compiler' - The New Hacker's Dictionary, 1996
- front-end [devs] business layer [backend devs] monolithic database [DBA team]
- a web api controller [manager] delegates most business logic to business classes [developers] which are part of the same in-memory process [team], while serving as a single entry-point for wider cross-network [cross-team] communication.

Conway's law: more examples

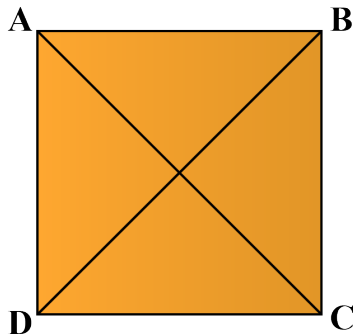
- separate .csproj projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations
- all business documentation in the same repository, separate from the repositories that the documentation is for

Conway's law: corollaries

- Conway's law can't be fought
- Teams working in the same part of the same codebase at the same time are the same team
 - separate the codebases for improved velocity
 - join the teams for improved code sharing and communication (Inverse conway maneuver)
 - branching-based strategies for team management compromise between the above approaches without their benefits

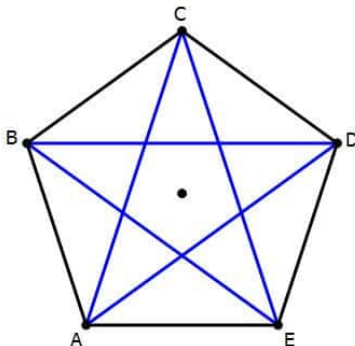
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $n(n-1)/2$



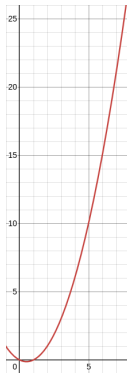
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- number of direct communication paths for n individuals= $\frac{n(n-1)}{2}$



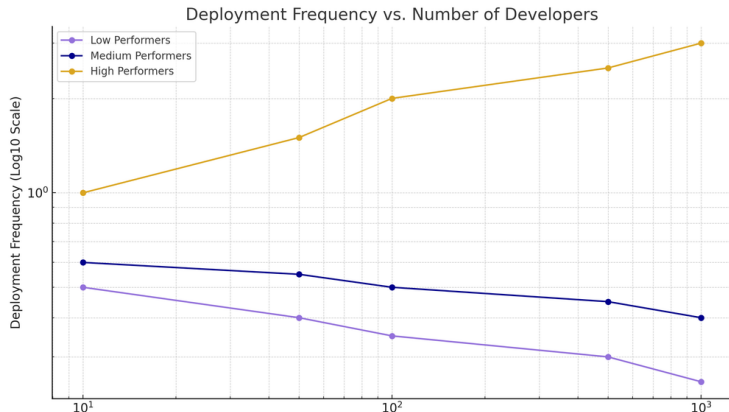
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- number of direct communication paths for n individuals=
$$\frac{n(n-1)}{2}$$



The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



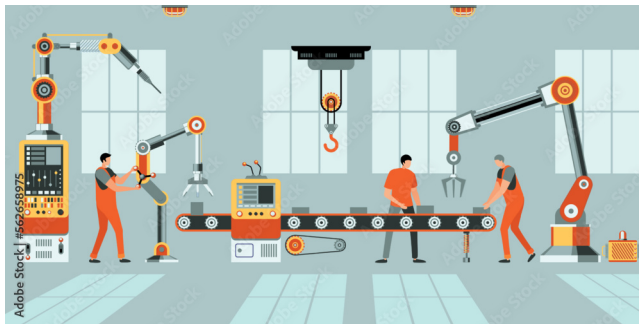
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Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
- Decrease:
 - operating costs
 - inventory
 - scrap
- Remove bottlenecks

Goldratt's theory of constraints (cont.)



2013: The State of DevOps Report

- a series of surveys of over 36,000 professionals
- Led by a PhD statistician, Nicole Forsgren, from 2013-2019
- Uses *cluster analysis* to discover groupings - has no prior understanding of what counts as good or bad.

The State of DevOps Report: throughput metrics

- lead time for changes
- deployment frequency

The State of DevOps Report: stability metrics

- change failure rate
- mean time to restore

The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x

faster lead
time

182x

more
deployments
per year

8x

lower change
failure rate

2293x

faster failed
deployment
recovery times

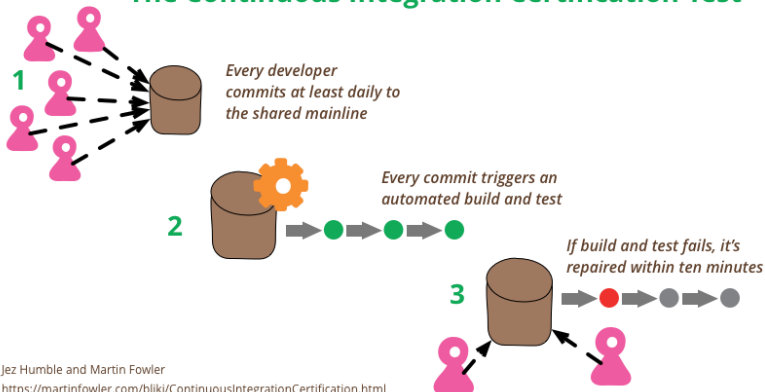
- 2024 State of DevOps Report

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Defining continuous integration

The Continuous Integration Certification Test



Jez Humble and Martin Fowler

<https://martinfowler.com/bliki/ContinuousIntegrationCertification.html>

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Continuous integration branching patterns

- don't
- as little as possible, for as short a time as possible, at least daily

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Continuous integration: trade-offs

Continuous Integration applies a different trigger for integration - you integrate whenever you've made a hunk of progress on the feature and your branch is still healthy. There's no expectation that the feature be complete, just that there's been a worthwhile amount of changes to the codebase. The rule of thumb is that "everyone commits to the mainline every day", or more precisely: you should never have more than a day's work sitting unintegrated in your local repository.

- Martin Fowler, "Branching Patterns"

Feedback comes late

Too late to change anything substantial

Low quality feedback

Through comments, lacking nuance

"My code"

Person owns area of code; key-person dependencies

Individual coding styles

Harder to have standards

Infrequent integration

Only integrate at the end of story/feature

Easy to ignore a failing build

Because it runs on an isolated branch

Dread large refactorings

Because they are likely to cause merge conflicts

People work in isolation

Harder to spot if someone needs help

Feedback comes in real time

As you're writing code or even before

Better quality feedback

Through face-to-face discussion

"Our code"

Collective code ownership

Team coding style

Easier to establish standards when pairing

Continuous Integration

Integrate as soon as code is pushed

We get used to not breaking things

Broken build on master is the team top priority

Easier to tackle large refactorings

As the rest of the team is always up to date

Visibility of what everyone is doing

Easier to spot if someone needs help

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Continuous integration: Enablers

- fast PR turnaround
- improving CI pipeline runtime
- team discipline

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Continuous integration: benefits

- High performers have the shortest integration times and branch lifetimes, with branch life and integration typically lasting hours.
- Low performers have the longest integration times and branch lifetimes, with branch life and integration typically lasting days.
- These differences are statistically significant
 - 2017 State of DevOps Report, p. 41

Continuous integration, benefits (cont.)

Taken together, continuous delivery and organizational culture explain 28.8% of the variance in IT performance.

- Forsgren and Humble (2016), 'The Role of Continuous Delivery in IT and Organizational Performance'

Continuous integration, benefits (cont.)

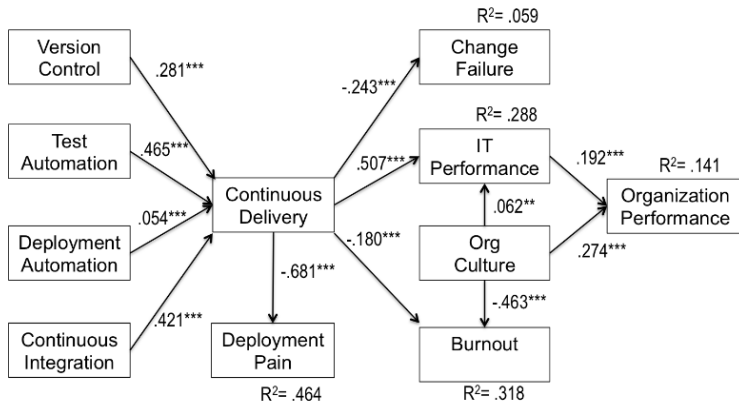


Figure 2. Emergent Model

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communication anti-pattern: local optimization

- example: separate (DevOps, QA, operations, support) team
- leads to bottlenecks
- delays communication
- discourages cross-functional collaboration
- punishes untracked work (e.g. thorough code reviews/dev testing, helping blocked colleagues)

Solution

- autonomous, cross-functional teams with 't-shaped' employees
- communities of practice

Revisiting Conway's law: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.
 - pull requests
 - testing
 - deployment

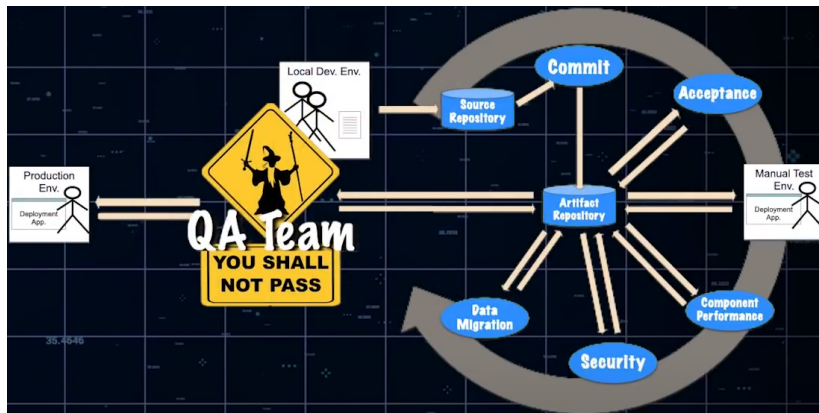
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Throughput anti-pattern: Gandalf vs. the Balrog



Throughput anti-pattern: Gandalf vs. the Balrog



Throughput anti-pattern: Gandalf vs. the Balrog

- change approval boards
- pre-merge code reviews
- 'bridge-to-nowhere' pipelines
- these all increase lead time

Throughput anti-pattern solutions: aggressively parallelize work

- pair programming
- parallelized pipeline builds for 'wide' pipelines
- one build artifact for all pipeline stages
- share responsibility

Defining throughput for our project

- Avoid 'thrashing' metrics
- *Throughput* - the total lapsed time from local commit to production deployment
- *Throughput** - the total lapsed time from local commit to user acceptance testing handoff

Conclusion

- Reduce goal misalignment
- stop people from waiting on each other
- work in small batches
- That's mostly it

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